

Connor Savugot

B.S. in Computer Engineering | North Carolina State University | Graduated May 2026

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EMBEDDED CONTROLS PROFILE

Embedded firmware and controls engineer developing C software for power and communications systems. Combines professional MATLAB data analysis, CAN/J1939, FreeRTOS, and hardware integration with academic MATLAB/Simulink Model-in-the-Loop, PID, robotics, and autonomous-vehicle controls.

RELEVANT SKILLS

Languages & Modeling: Embedded C, C++, MATLAB, Simulink, Python, Lustre

Controls & Real-Time: Model-Based Design, Model-in-the-Loop, PID/PI/PD tuning, cascaded and discrete-time control, state-space/transfer-function analysis

Platforms & Operating Systems: FreeRTOS, bare metal, Embedded Linux; dsPIC33CK, PIC32, TMS320 DSP, STM32G474; interrupts/timers, peripheral/HAL driver bring-up

Vehicle Networks & I/O: Classical CAN (CAN 2.0), CAN FD, J1939; ADC, PWM, I2C, SPI, UART/RS-232

Development, Delivery & Validation: Requirements analysis, interface specifications, software integration, unit, component, and functional testing, design reviews, Git/GitLab, GCC/Make build automation, Code Composer Studio, JTAG, oscilloscopes, logic/protocol analyzers, schematics/datasheets

ENGINEERING EXPERIENCE

AEGIS Power Systems

Embedded Firmware Engineer

May 2026–Present

- Built a MATLAB workflow to parse captured CAN telemetry and compare candidate digital filters by reading variation under steady input and time to follow large electrical changes; selected rolling-average and EMA approaches, derived per-channel scaling, and implemented rolling-sum averaging in embedded C.
- Develop low-level C firmware for 16-bit dsPIC33CK DSCs and bring up peripherals from schematics, datasheets, reference manuals, and protocol specifications; isolate register, timing, configuration, and electrical faults with JTAG, targeted tests, oscilloscopes, and logic analyzers.
- Integrate CAN-connected power hardware with TI AM62 Embedded Linux software, tracing defects across wiring, bus traffic, device configuration, services, and application code; coordinate priorities, findings, and validation through standups with a distributed firmware team.

Embedded Systems Intern

May–Aug 2025

Computer Science Intern

May–Aug 2024

Selected work across both internships

- Developed bare-metal C power-control firmware for a TMS320 DSP driving transformer-coupled bidirectional buck-boost supplies; coordinated switching across 12 PWM A/B pairs (24 outputs) and validated ADC/PWM behavior in Code Composer Studio.
- Developed embedded C firmware for a PIC32 running FreeRTOS, coordinating CAN/J1939, I2C, SPI, UART, and update behavior within a two-processor PIC32/TMS320 product; built a multistage update workflow using C++, Python, and embedded firmware to transfer, program, verify, and recover firmware images.

TEAM Industries

Engineering Intern

Jun–Aug 2022 and Jun–Aug 2021

- Built a repeatable Excel/VBA dimensional and spline-tolerance validation workflow that flagged out-of-spec parts; maintained structured quality records and collaborated with engineering and manufacturing teams on test procedures and revision tracking.

CONTROLS & VEHICLE SOFTWARE PROJECTS

Electric Zero-Turn Mower Controls | MATLAB/Simulink

Academic Controls Project

- Developed MATLAB/Simulink Model-in-the-Loop tooling to tune PID gains and exercise control logic; created a repeatable gain-adjustment method and compared simulated response with real-vehicle test behavior.

Robotics & Digital Controls | Python

Fall 2025

- Extended Python simulations for mass-spring-damper, planar-VTOL, and Hummingbird models; derived linearized/state-space models and tuned cascaded PD/PID controllers with saturation, anti-windup, and PWM actuator mixing.

Embedded Systems for Autonomous Driving | Lustre, Python/CARLA

Spring 2026

- Implemented and tuned a deterministic PID cruise controller in Lustre with initialized state, discrete-time integral/derivative terms, and output limits; compared its simulated response with an earlier controller. Separately used Luke counterexample traces to correct temporal monitors and verify state-machine properties.
- On a three-person team, integrated a Python agent with CARLA's planning stack for PID-based longitudinal control; tuned target-speed response and evaluated Town10HD runs using completion time, collisions, and lane invasions.

EV Active Sensor Adapter | Embedded C, CAN, Requirements & Test

Senior Design

- Translated battery-sensing and CAN constraints into system/interface requirements and validation criteria for a team-built STM32G474/BQ79616 adapter; supported design reviews and staged integration testing, including seven passing host-side tests for conversions, CAN payloads, boundary cases, and checksums.